# Results

## Descriptives

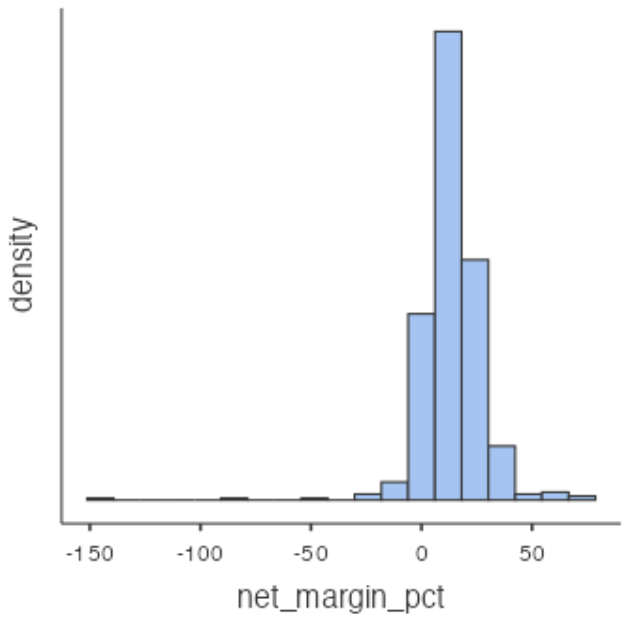
Descriptives

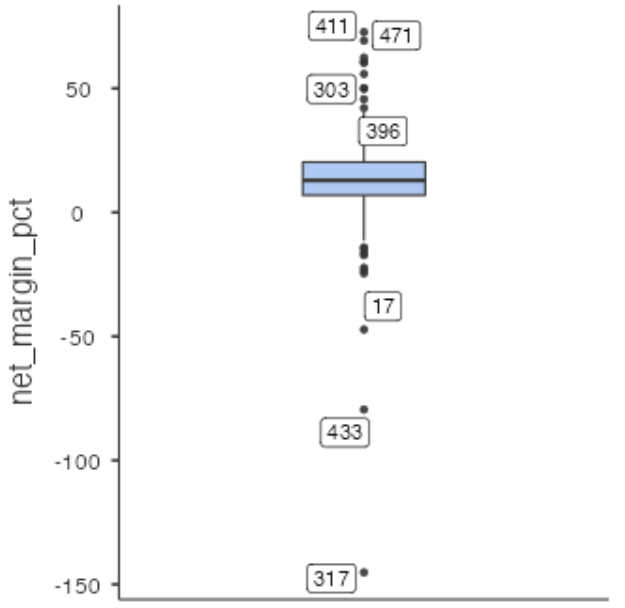
| | net_margin_pct | debt_ratio_pct | revenue_growth_pct |
|---|---|---|---|
| N | 498 | 498 | 488 |
| Missing | 2 | 2 | 12 |
| Mean | 13.8 | 66.7 | 8.64 |
| Median | 12.9 | 64.2 | 6.37 |
| Mode | -145[a] | 44.5[a] | 3.67[a] |
| Standard deviation | 14.7 | 28.0 | 15.1 |
| Minimum | -145 | 8.96 | -42.4 |
| Maximum | 72.7 | 327 | 186 |
| Skewness | -2.67 | 2.76 | 4.76 |
| Std. error skewness | 0.109 | 0.109 | 0.111 |
| Kurtosis | 31.3 | 20.2 | 45.1 |
| Std. error kurtosis | 0.218 | 0.218 | 0.221 |
| Shapiro-Wilk W | 0.791 | 0.837 | 0.661 |
| Shapiro-Wilk p | <.001 | <.001 | <.001 |
| 25th percentile | 6.93 | 49.9 | 2.11 |
| 50th percentile | 12.9 | 64.2 | 6.37 |
| 75th percentile | 20.3 | 78.6 | 11.0 |

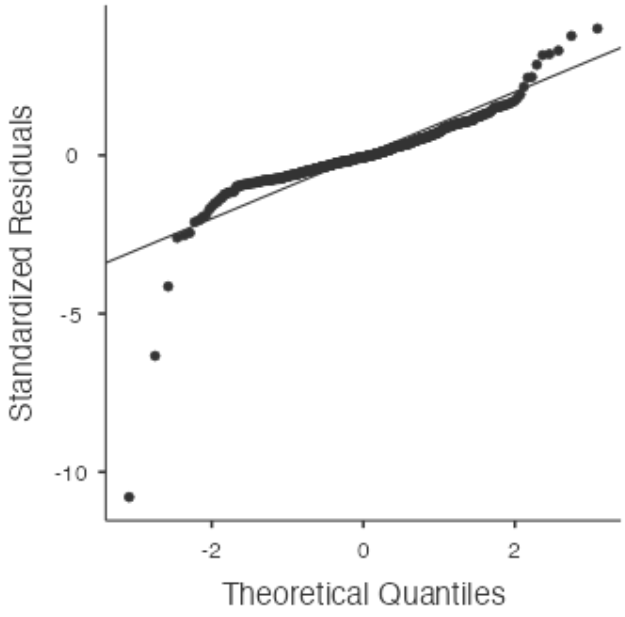
[a] More than one mode exists, only the first is reported

## Plots

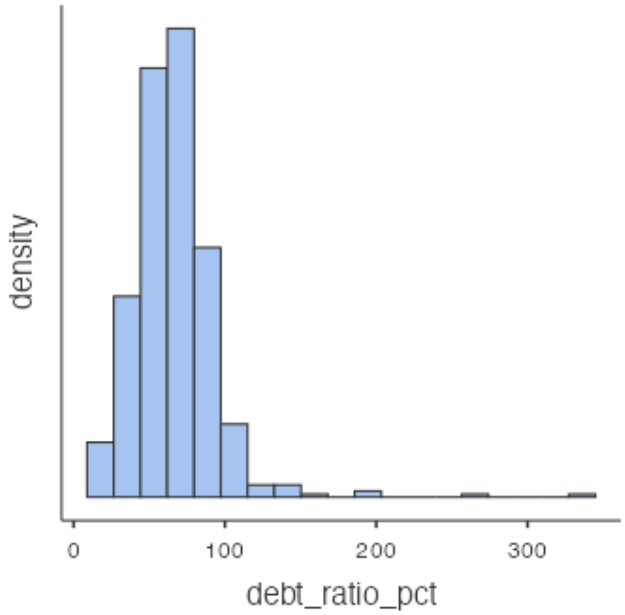
**net_margin_pct**

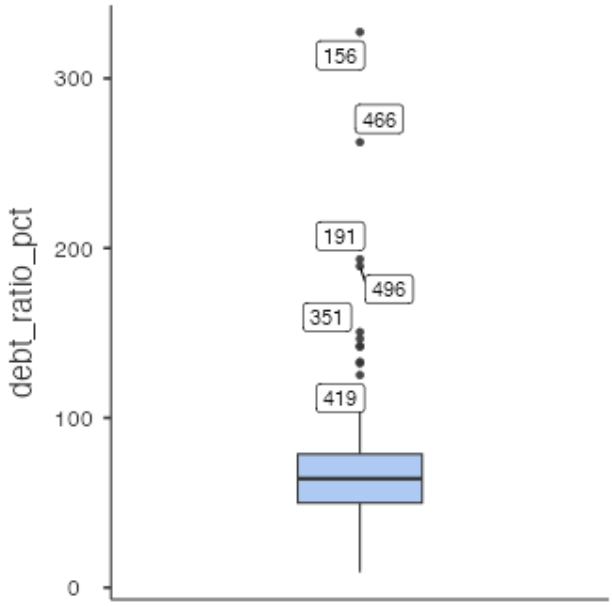


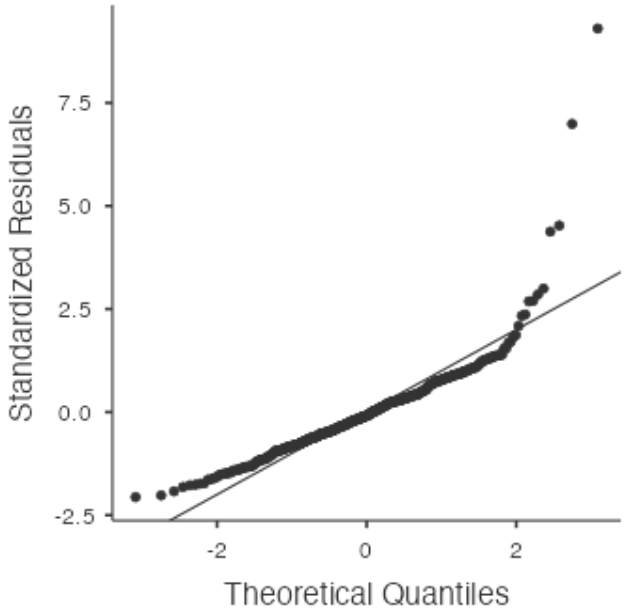


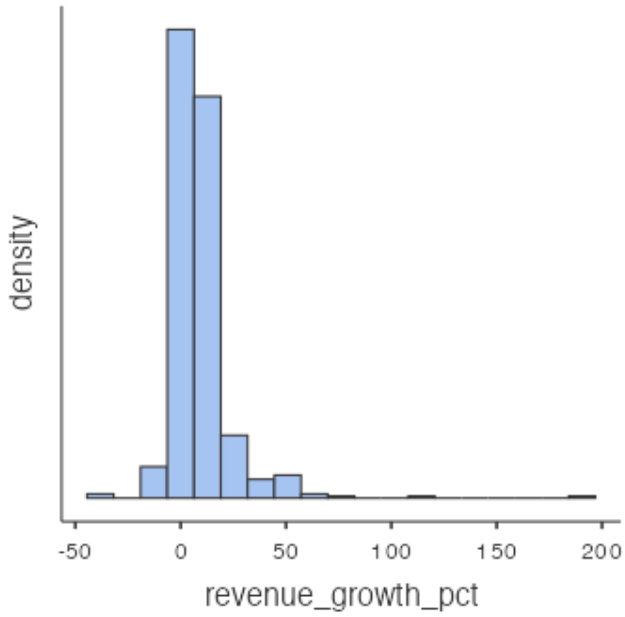

**debt_ratio_pct**

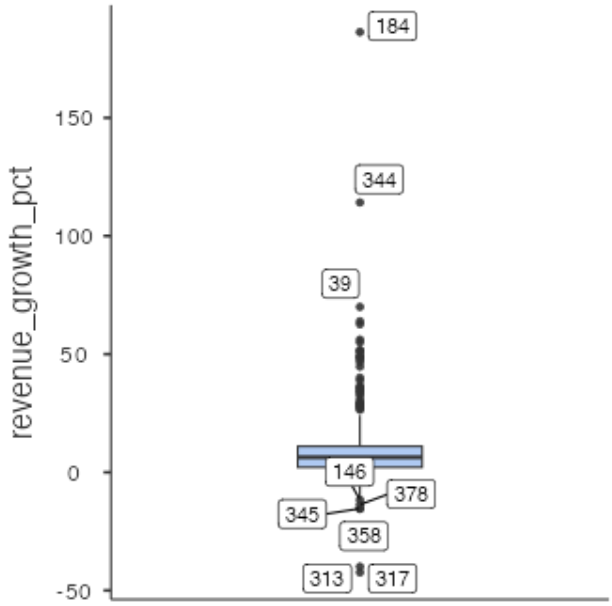

density
0
100
200
300
debt_ratio_pct

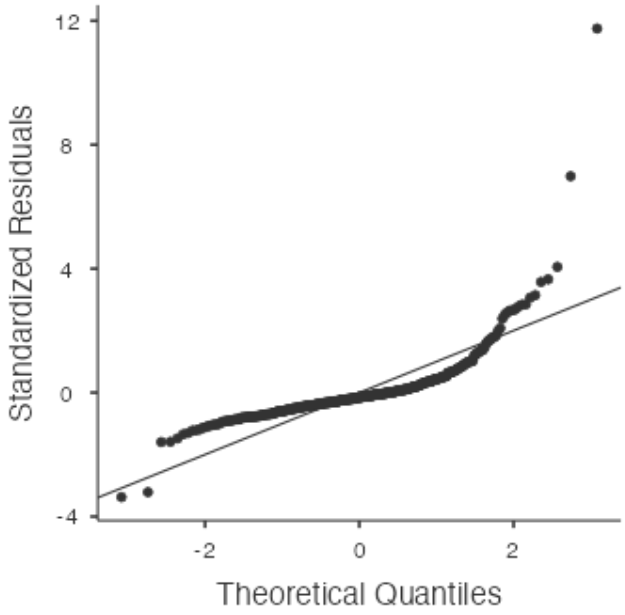

debt_ratio_pct
300
200
100
0
156
466
191
496
351
419




**revenue_growth_pct**






## Descriptives

Descriptives

| | sector_group | size_quartile |
|---|---|---|
| N | 500 | 498 |
| Missing | 0 | 2 |

### Frequencies

Frequencies of sector_group

| sector_group | Counts | % of Total | Cumulative % |
|---|---|---|---|
| Consumer | 115 | 23.0% | 23.0% |
| Finance | 76 | 15.2% | 38.2% |

Frequencies of sector_group

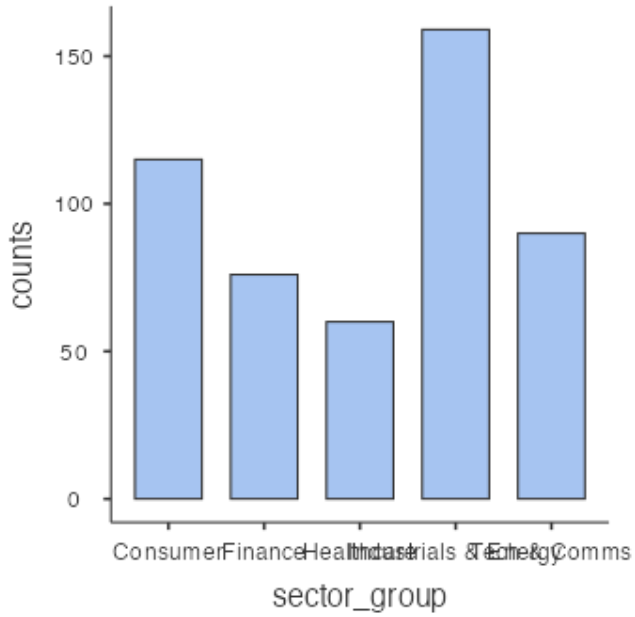
| | | | |
|---|---|---|---|
| Healthcare | 60 | 12.0% | 50.2% |
| Industrials & Energy | 159 | 31.8% | 82.0% |
| Tech & Comms | 90 | 18.0% | 100.0% |

Frequencies of size_quartile

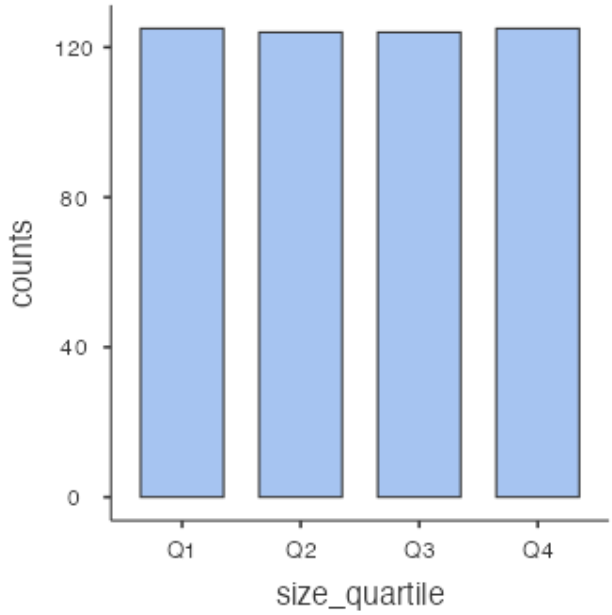
| size_quartile | Counts | % of Total | Cumulative % |
|---|---|---|---|
| Q1 | 125 | 25.1% | 25.1% |
| Q2 | 124 | 24.9% | 50.0% |
| Q3 | 124 | 24.9% | 74.9% |
| Q4 | 125 | 25.1% | 100.0% |

## Plots

**sector_group**




**size_quartile**




## Contingency Tables

Contingency Tables

| sector_group | | size_quartile Q1 | Q2 | Q3 | Q4 | Total |
|---|---|---|---|---|---|---|
| Consumer | Observed | 33 | 40 | 30 | 12 | 115 |
| | Expected | 28.9 | 28.6 | 28.6 | 28.9 | 115.0 |
| | % within row | 28.7% | 34.8% | 26.1% | 10.4% | 100.0% |
| Finance | Observed | 5 | 10 | 15 | 46 | 76 |
| | Expected | 19.1 | 18.9 | 18.9 | 19.1 | 76.0 |
| | % within row | 6.6% | 13.2% | 19.7% | 60.5% | 100.0% |
| Healthcare | Observed | 18 | 14 | 10 | 18 | 60 |
| | Expected | 15.1 | 14.9 | 14.9 | 15.1 | 60.0 |
| | % within row | 30.0% | 23.3% | 16.7% | 30.0% | 100.0% |
| Industrials & Energy | Observed | 38 | 36 | 57 | 27 | 158 |
| | Expected | 39.7 | 39.3 | 39.3 | 39.7 | 158.0 |
| | % within row | 24.1% | 22.8% | 36.1% | 17.1% | 100.0% |
| Tech & Comms | Observed | 31 | 24 | 12 | 22 | 89 |
| | Expected | 22.3 | 22.2 | 22.2 | 22.3 | 89.0 |
| | % within row | 34.8% | 27.0% | 13.5% | 24.7% | 100.0% |
| Total | Observed | 125 | 124 | 124 | 125 | 498 |
| | Expected | 125 | 124 | 124 | 125 | 498 |
| | % within row | 25.1% | 24.9% | 24.9% | 25.1% | 100.0% |

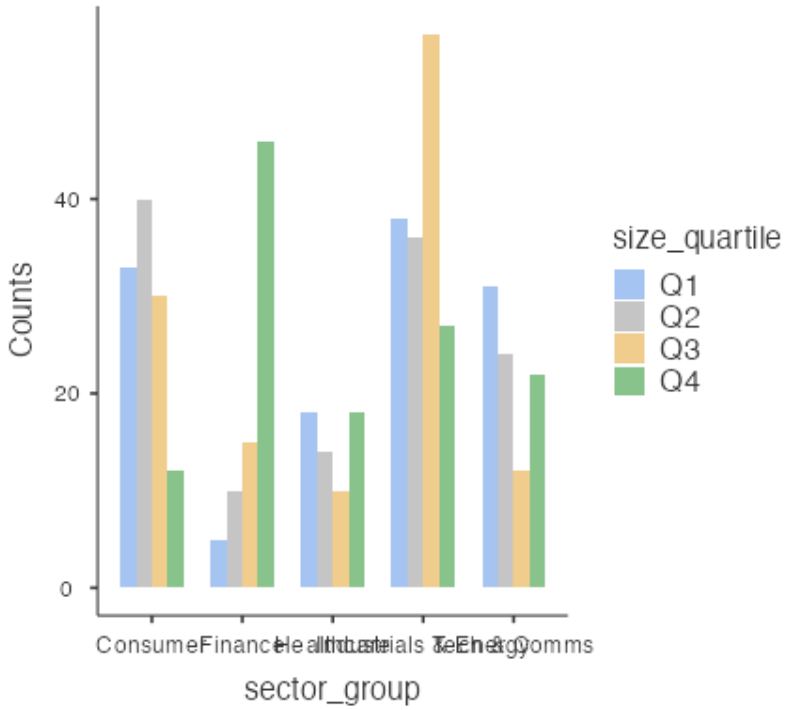
$\chi^2$ Tests

| | Value | df | p |
|---|---|---|---|
| $\chi^2$ | 91.8 | 12 | <.001 |
| N | 498 | | |

Nominal

| | Value |
|---|---|
| Phi-coefficient | NaN |
| Cramer's V | 0.248 |

## Plots



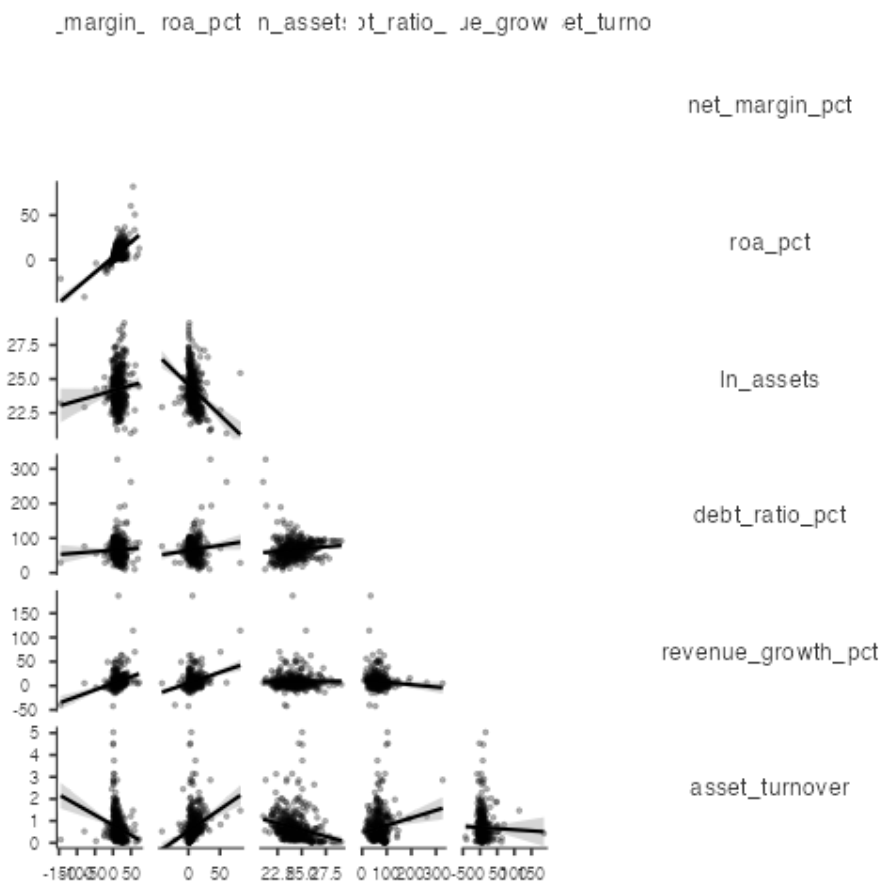

## Correlation Matrix

Correlation Matrix

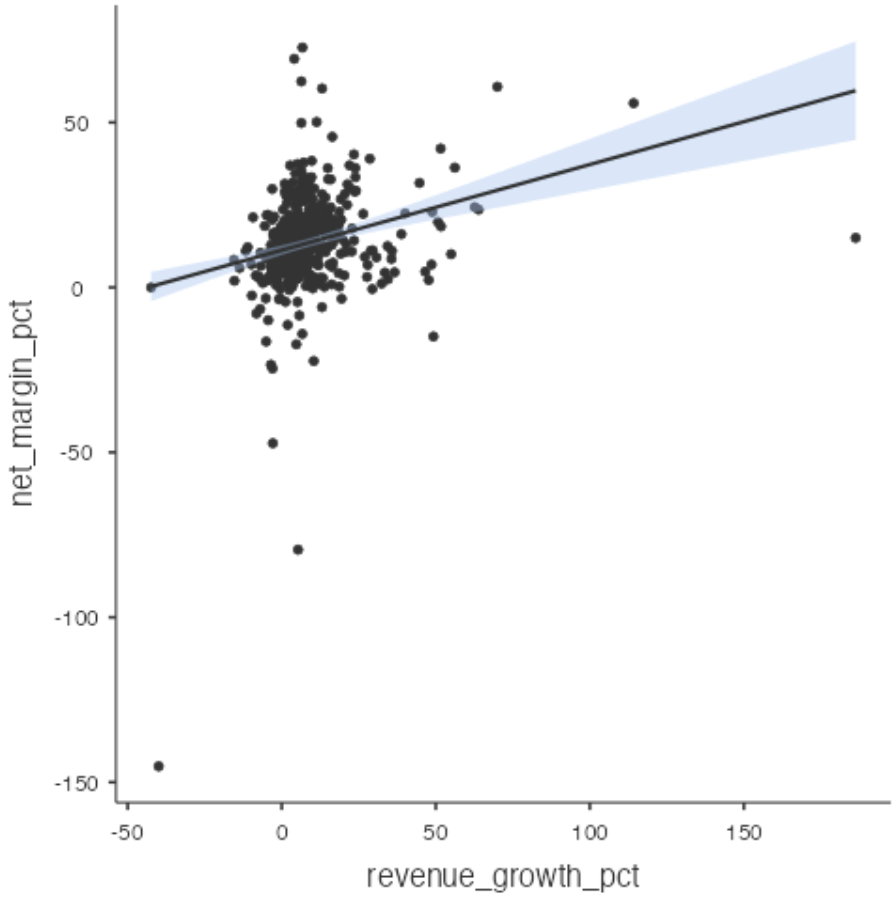
| | | net_margin_pct | roa_pct | ln_assets | debt_ratio_pct | revenue_growth_pct | asset_turnover |
|---|---|---|---|---|---|---|---|
| net_margin_pct | Pearson's r | — | | | | | |
| | df | — | | | | | |
| | p-value | — | | | | | |
| | Spearman's rho | — | | | | | |
| | df | — | | | | | |
| | p-value | — | | | | | |
| roa_pct | Pearson's r | 0.582*** | — | | | | |
| | df | 496 | — | | | | |
| | p-value | <.001 | — | | | | |
| | Spearman's rho | 0.487*** | — | | | | |
| | df | 496 | — | | | | |
| | p-value | <.001 | — | | | | |
| ln_assets | Pearson's r | 0.085 | -0.295*** | — | | | |
| | df | 496 | 496 | — | | | |
| | p-value | .059 | <.001 | — | | | |
| | Spearman's rho | 0.057 | -0.377*** | — | | | |
| | df | 496 | 496 | — | | | |
| | p-value | .204 | <.001 | — | | | |
| debt_ratio_pct | Pearson's r | 0.044 | 0.089* | 0.119** | — | | |
| | df | 496 | 496 | 496 | — | | |
| | p-value | .332 | .046 | .008 | — | | |
| | Spearman's rho | -0.059 | -0.173*** | 0.293*** | — | | |
| | df | 496 | 496 | 496 | — | | |
| | p-value | .190 | <.001 | <.001 | — | | |
| revenue_growth_pct | Pearson's r | 0.266*** | 0.258*** | 0.008 | -0.090* | — | |
| | df | 486 | 486 | 486 | 486 | — | |
| | p-value | <.001 | <.001 | .865 | .046 | — | |
| | Spearman's rho | 0.295*** | 0.137** | 0.013 | -0.066 | — | |
| | df | 486 | 486 | 486 | 486 | — | |
| | p-value | <.001 | .002 | .782 | .143 | — | |
| asset_turnover | Pearson's r | -0.217*** | 0.272*** | -0.249*** | 0.153*** | -0.025 | — |
| | df | 496 | 496 | 496 | 496 | 486 | — |
| | p-value | <.001 | <.001 | <.001 | <.001 | .581 | — |
| | Spearman's rho | -0.375*** | 0.524*** | -0.402*** | -0.035 | -0.075 | — |
| | df | 496 | 496 | 496 | 496 | 486 | — |
| | p-value | <.001 | <.001 | <.001 | .436 | .096 | — |

Note. * p < .05, ** p < .01, *** p < .001

## Plot




## Scatter Plot




## Independent Samples T-Test

Independent Samples T-Test

| | | | | | | | 95% Confidence Interval | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Statistic | df | p | Mean difference | SE difference | Lower | Upper | | Effect Size |
| net_margin_pct | Student's t | 4.70 | 496 | <.001 | 8.44 | 1.80 | 4.91 | 12.0 | Cohen's d | 0.585 |
| | Welch's t | 5.70 | 128 | <.001 | 8.44 | 1.48 | 5.51 | 11.4 | Cohen's d | 0.639 |
| | Mann-Whitney U | 9327 | | <.001 | 7.93 | | 5.34 | 10.4 | Rank biserial correlation | -0.418 |

Note. $H_a$ $\mu_{Finance} \neq \mu_{Non\text{-}finance}$

### Assumptions

Normality Test (Shapiro-Wilk)

| | W | p |
|---|---|---|
| net_margin_pct | 0.781 | <.001 |

Note. A low p-value suggests a violation of the assumption of normality

Homogeneity of Variances Test (Levene's)

| | F | df | df2 | p |
|---|---|---|---|---|
| net_margin_pct | 0.117 | 1 | 496 | .733 |

Note. A low p-value suggests a violation of the assumption of equal variances

**[4]**

Group Descriptives

| | Group | N | Mean | Median | SD | SE |
|---|---|---|---|---|---|---|
| net_margin_pct | Finance | 76 | 20.9 | 21.0 | 11.2 | 1.29 |
| | Non-finance | 422 | 12.5 | 12.1 | 14.9 | 0.726 |

## One-Way ANOVA

One-Way ANOVA

| | | F | df1 | df2 | p |
|---|---|---|---|---|---|
| net_margin_pct | Welch's | 10.4 | 4 | 189 | <.001 |
| | Fisher's | 7.52 | 4 | 493 | <.001 |

Group Descriptives

| | sector_group | N | Mean | SD | SE |
|---|---|---|---|---|---|
| net_margin_pct | Consumer | 115 | 11.92 | 15.02 | 1.401 |
| | Finance | 76 | 20.90 | 11.24 | 1.289 |
| | Healthcare | 60 | 9.51 | 23.10 | 2.983 |
| | Industrials & Energy | 158 | 12.07 | 8.62 | 0.686 |
| | Tech & Comms | 89 | 15.87 | 16.22 | 1.719 |

### Assumption Checks

Normality Test (Shapiro-Wilk)

| | W | p |
|---|---|---|
| net_margin_pct | 0.781 | <.001 |

Note. A low p-value suggests a violation of the assumption of normality

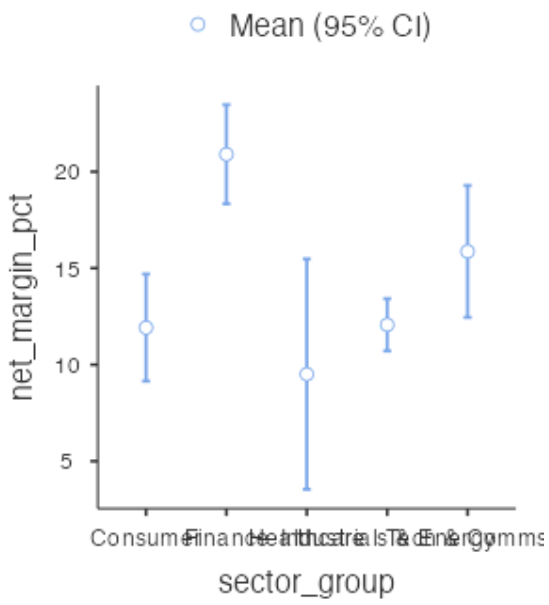
Homogeneity of Variances Test (Levene's)

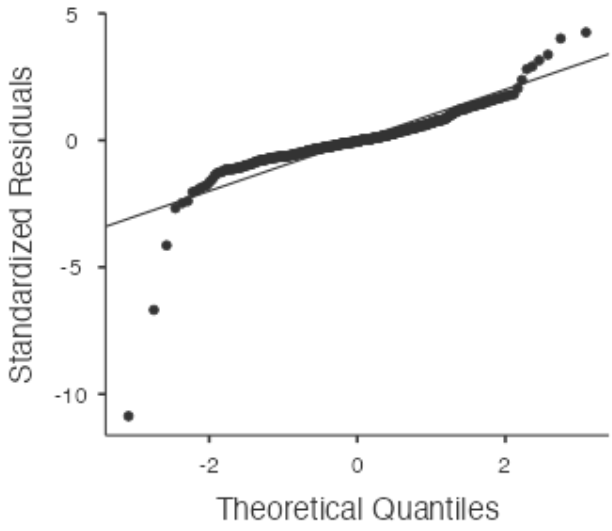
| | F | df1 | df2 | p |
|---|---|---|---|---|
| net_margin_pct | 3.61 | 4 | 493 | .007 |

**[4]**

### Plots

**net_margin_pct**




## Post Hoc Tests

Games-Howell Post-Hoc Test – net_margin_pct

| | | Consumer | Finance | Healthcare | Industrials & Energy | Tech & Comms |
|---|---|---|---|---|---|---|
| Consumer | Mean difference | — | -8.98*** | 2.41 | -0.143 | -3.94 |
| | p-value | — | <.001 | .948 | 1.000 | .389 |
| Finance | Mean difference | | — | 11.39** | 8.837*** | 5.04 |
| | p-value | | — | .007 | <.001 | .137 |
| Healthcare | Mean difference | | | — | -2.555 | -6.36 |
| | p-value | | | — | .919 | .353 |
| Industrials & Energy | Mean difference | | | | — | -3.80 |
| | p-value | | | | — | .248 |
| Tech & Comms | Mean difference | | | | | — |
| | p-value | | | | | — |

Note. * p < .05, ** p < .01, *** p < .001

## ANOVA

ANOVA - net_margin_pct

| | Sum of Squares | df | Mean Square | F | p | η² |
|---|---|---|---|---|---|---|
| sector_group | 6198 | 4 | 1549 | 7.52 | <.001 | 0.058 |
| Residuals | 101519 | 493 | 206 | | | |

**[4]**

## One-Way ANOVA (Non-parametric)

Kruskal-Wallis

| | χ² | df | p | ε² |
|---|---|---|---|---|
| net_margin_pct | 46.2 | 4 | <.001 | 0.0930 |

### Dwass-Steel-Critchlow-Fligner pairwise comparisons

Pairwise comparisons - net_margin_pct

| | | W | p |
|---|---|---|---|
| Consumer | Finance | 7.416 | <.001 |
| Consumer | Healthcare | 0.413 | .998 |
| Consumer | Industrials & Energy | 1.636 | .776 |
| Consumer | Tech & Comms | 4.267 | .022 |
| Finance | Healthcare | -6.006 | <.001 |
| Finance | Industrials & Energy | -8.731 | <.001 |
| Finance | Tech & Comms | -3.629 | .077 |
| Healthcare | Industrials & Energy | 0.364 | .999 |
| Healthcare | Tech & Comms | 3.240 | .148 |
| Industrials & Energy | Tech & Comms | 4.468 | .014 |

# Linear Regression

Model Fit Measures

| | | | | Overall Model Test | | | |
|---|---|---|---|---|---|---|---|
| Model | R | $R^2$ | Adjusted $R^2$ | F | df1 | df2 | p |
| 1 | 0.0848 | 0.00719 | 0.00519 | 3.59 | 1 | 496 | .059 |

Note. Models estimated using sample size of N=498

Model Coefficients - net_margin_pct

| | | | 95% Confidence Interval | | | | |
|---|---|---|---|---|---|---|---|
| Predictor | Estimate | SE | Lower | Upper | t | p | Stand. Estimate |
| Intercept | -9.489 | 12.279 | -33.6152 | 14.64 | -0.773 | .440 | |
| ln_assets | 0.958 | 0.506 | -0.0350 | 1.95 | 1.895 | .059 | 0.0848 |

# Linear Regression

Model Fit Measures

| | | | | Overall Model Test | | | |
|---|---|---|---|---|---|---|---|
| Model | R | $R^2$ | Adjusted $R^2$ | F | df1 | df2 | p |
| 1 | 0.337 | 0.114 | 0.101 | 8.80 | 7 | 480 | <.001 |

Note. Models estimated using sample size of N=488

Model Coefficients - net_margin_pct

| | | | 95% Confidence Interval | | | | |
|---|---|---|---|---|---|---|---|
| Predictor | Estimate | SE | Lower | Upper | t | p | Stand. Estimate |
| Intercept[a] | 8.5739 | 12.8261 | -16.6283 | 33.7761 | 0.6685 | .504 | |
| debt_ratio_pct | 0.0177 | 0.0232 | -0.0280 | 0.0633 | 0.7603 | .447 | 0.03371 |
| revenue_growth_pct | 0.2482 | 0.0429 | 0.1639 | 0.3325 | 5.7835 | <.001 | 0.25440 |
| ln_assets | 0.0119 | 0.5302 | -1.0298 | 1.0536 | 0.0225 | .982 | 0.00105 |
| sector_group: | | | | | | | |
| Consumer – Industrials & Energy | 0.2412 | 1.7519 | -3.2012 | 3.6836 | 0.1377 | .891 | 0.01636 |
| Finance – Industrials & Energy | 7.8380 | 2.1436 | 3.6260 | 12.0499 | 3.6565 | <.001 | 0.53145 |
| Healthcare – Industrials & Energy | -2.2083 | 2.1289 | -6.3914 | 1.9749 | -1.0373 | .300 | -0.14973 |
| Tech & Comms – Industrials & Energy | 2.3960 | 1.8752 | -1.2886 | 6.0806 | 1.2777 | .202 | 0.16246 |

[a] Represents reference level

**Data Summary**

Cook's Distance

| | | | Range | |
|---|---|---|---|---|
| Mean | Median | SD | Min | Max |
| 0.00439 | 3.14e-4 | 0.0406 | 8.88e-9 | 0.659 |

### Assumption Checks

Durbin–Watson Test for Autocorrelation

| Autocorrelation | DW Statistic | p |
|---|---|---|
| 0.0183 | 1.96 | .656 |

**[4]**

Collinearity Statistics

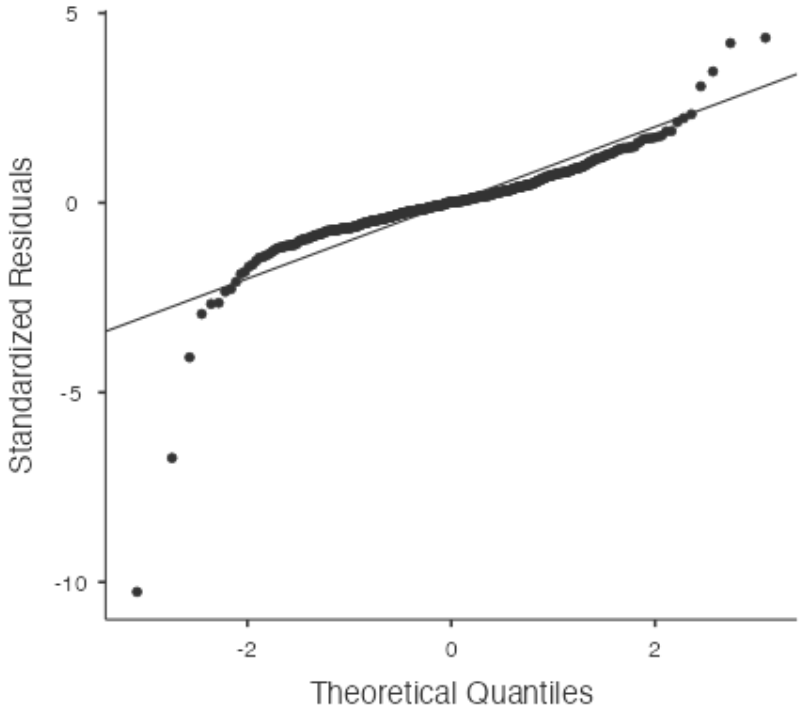
| | VIF | Tolerance |
|---|---|---|
| debt_ratio_pct | 1.03 | 0.969 |
| revenue_growth_pct | 1.02 | 0.977 |
| ln_assets | 1.08 | 0.922 |
| sector_group | 1.03 | 0.970 |

**[4]**

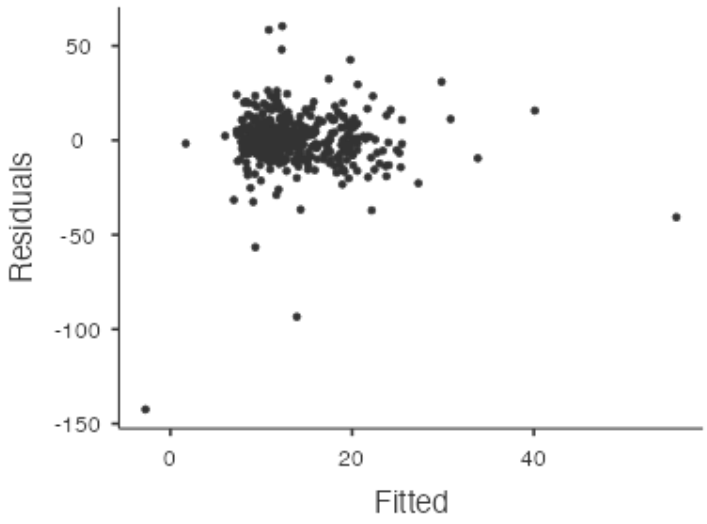
Normality Test (Shapiro-Wilk)

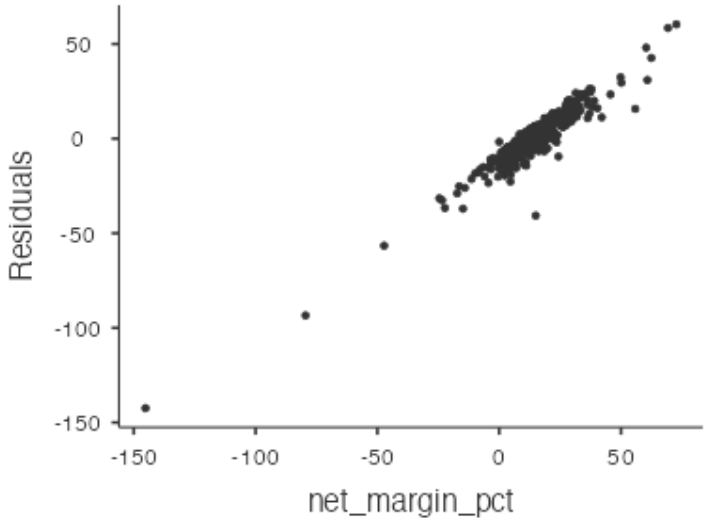
| Statistic | p |
|---|---|
| 0.798 | <.001 |

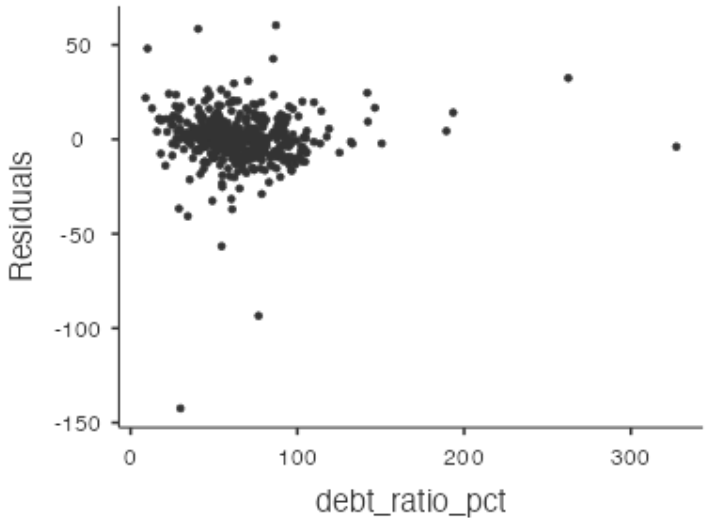
### Q-Q Plot

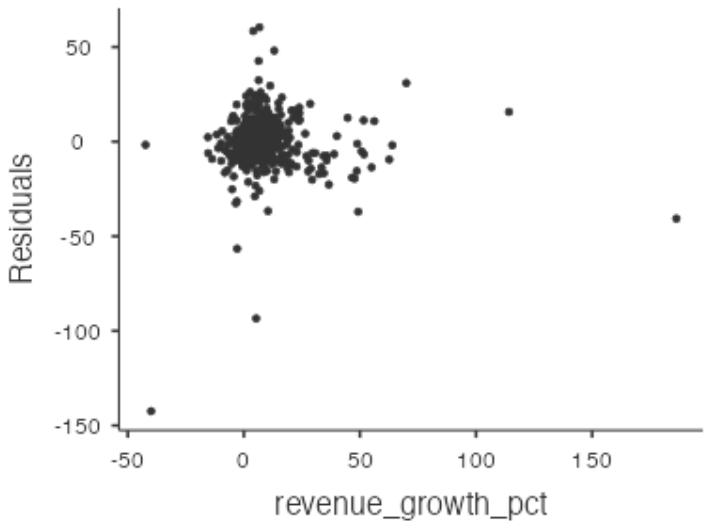


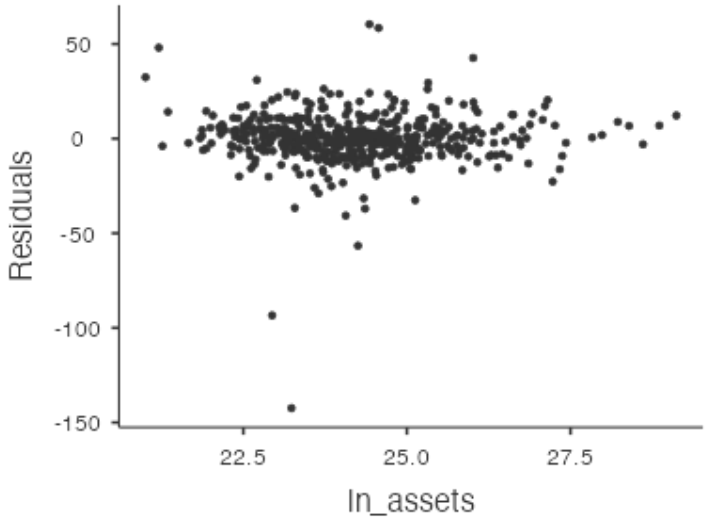

**Residuals Plots**




Residuals
50
0
-50
-100
-150
-150
-100
-50
0
50
net_margin_pct


Residuals
50
0
-50
-100
-150
0
100
200
300
debt_ratio_pct


Residuals
50
0
-50
-100
-150
-50
0
50
100
150
revenue_growth_pct




## Linear Regression

Model Fit Measures

| | | | | Overall Model Test | | | |
|---|---|---|---|---|---|---|---|
| Model | R | $R^2$ | Adjusted $R^2$ | F | df1 | df2 | p |
| 1 | 0.436 | 0.190 | 0.178 | 16.1 | 7 | 480 | <.001 |

Note. Models estimated using sample size of N=488

Model Coefficients - roa_pct

| | | | 95% Confidence Interval | | | | |
|---|---|---|---|---|---|---|---|
| Predictor | Estimate | SE | Lower | Upper | t | p | Stand. Estimate |
| Intercept[a] | 49.9287 | 7.2095 | 35.7627 | 64.0947 | 6.9254 | <.001 | |
| debt_ratio_pct | 0.0503 | 0.0131 | 0.0247 | 0.0760 | 3.8562 | <.001 | 0.16346 |
| revenue_growth_pct | 0.1449 | 0.0241 | 0.0976 | 0.1923 | 6.0092 | <.001 | 0.25272 |
| ln_assets | -1.9670 | 0.2980 | -2.5526 | -1.3815 | -6.6008 | <.001 | -0.29414 |
| sector_group: | | | | | | | |
| Consumer – Industrials & Energy | 0.1044 | 0.9847 | -1.8306 | 2.0393 | 0.1060 | .916 | 0.01204 |
| Finance – Industrials & Energy | -0.0816 | 1.2049 | -2.4491 | 2.2859 | -0.0677 | .946 | -0.00941 |
| Healthcare – Industrials & Energy | 0.5899 | 1.1967 | -1.7614 | 2.9412 | 0.4930 | .622 | 0.06803 |
| Tech & Comms – Industrials & Energy | 2.9983 | 1.0540 | 0.9272 | 5.0694 | 2.8446 | .005 | 0.34580 |

[a] Represents reference level

## Data Summary

Cook's Distance

| | | | Range | |
|---|---|---|---|---|
| Mean | Median | SD | Min | Max |
| 0.00672 | 3.21e-4 | 0.0627 | 2.90e-9 | 1.04 |

## Assumption Checks

Durbin–Watson Test for Autocorrelation

| Autocorrelation | DW Statistic | p |
|---|---|---|
| 0.113 | 1.77 | .010 |

**[4]**

Collinearity Statistics

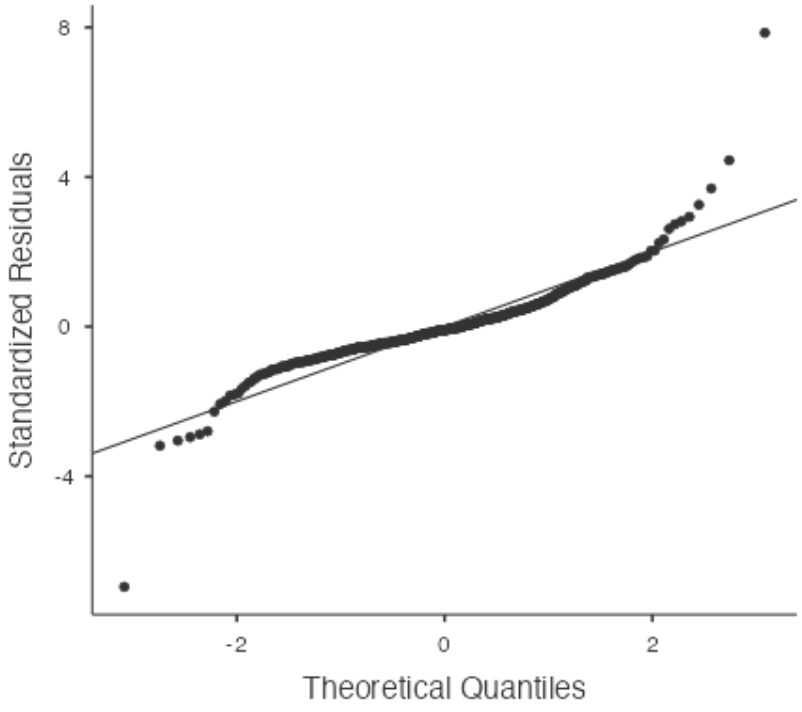
| | VIF | Tolerance |
|---|---|---|
| debt_ratio_pct | 1.03 | 0.969 |
| revenue_growth_pct | 1.02 | 0.977 |
| ln_assets | 1.08 | 0.922 |
| sector_group | 1.03 | 0.970 |

**[4]**

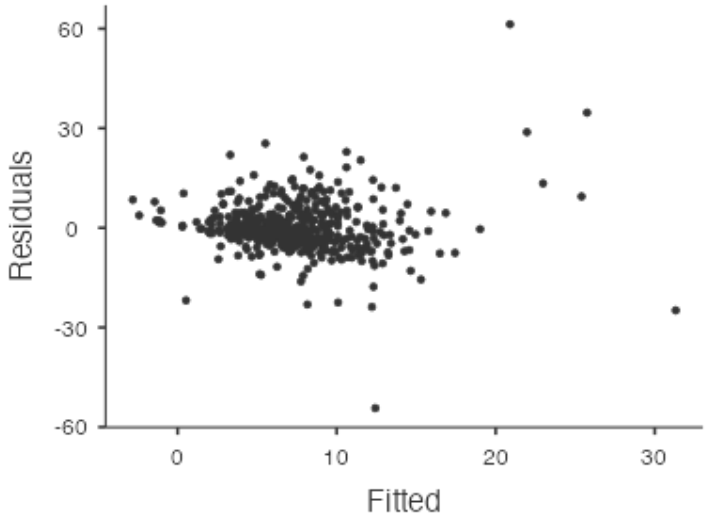
Normality Test (Shapiro-Wilk)

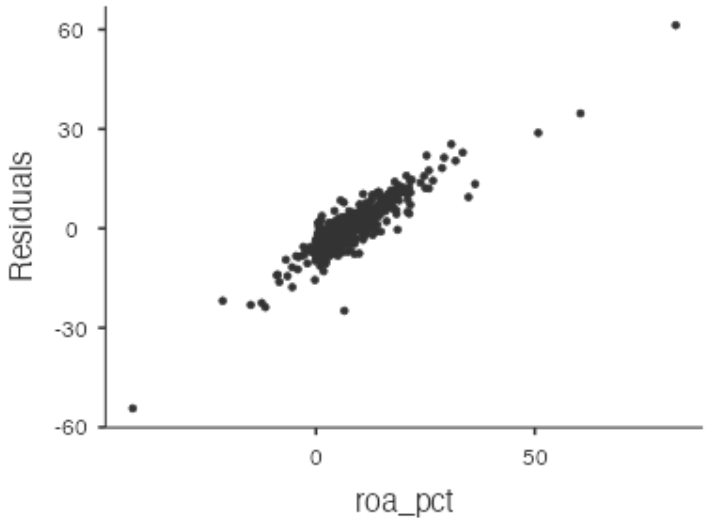
| Statistic | p |
|---|---|
| 0.866 | <.001 |

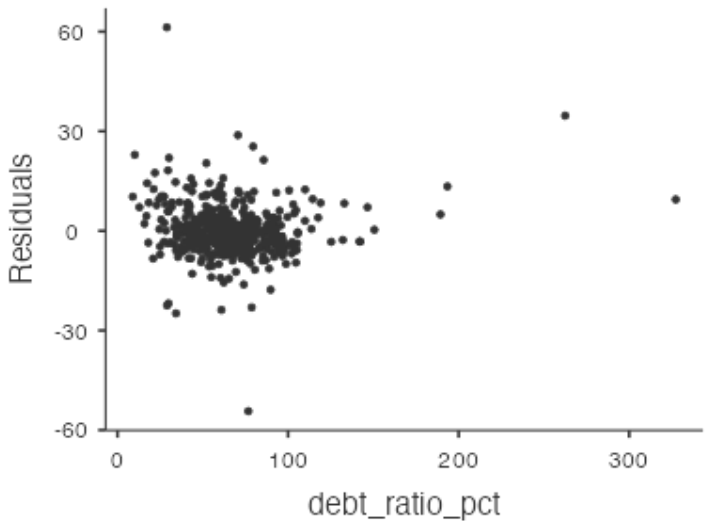
**Q-Q Plot**

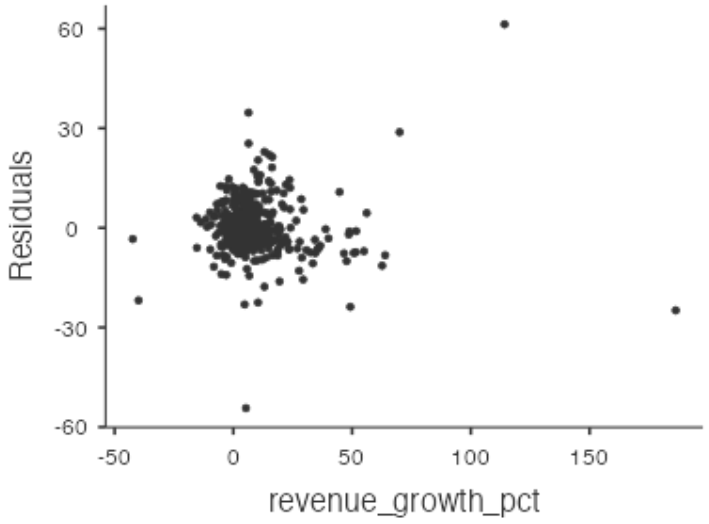


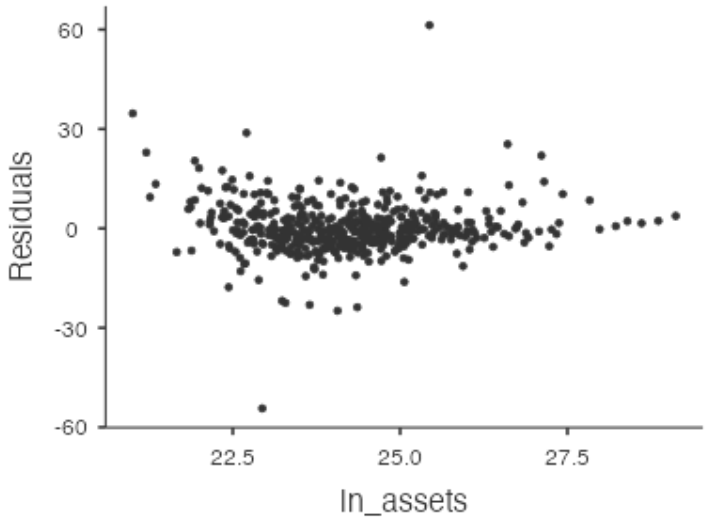

**Residuals Plots**


Residuals
60
30
0
-30
-60
0
10
20
30
Fitted


Residuals
60
30
0
-30
-60
0
50
roa_pct


Residuals
60
30
0
-30
-60
0
100
200
300
debt_ratio_pct




## Linear Regression

Model Fit Measures

| | | | | Overall Model Test | | | |
|---|---|---|---|---|---|---|---|
| Model | R | R² | Adjusted R² | F | df1 | df2 | p |
| 1 | 0.376 | 0.142 | 0.129 | 10.8 | 7 | 460 | <.001 |

Note. Models estimated using sample size of N=468

Model Coefficients - margin_noinfl

| Predictor | Estimate | SE | 95% Confidence Interval | | t | p | Stand. Estimate |
|---|---|---|---|---|---|---|---|
| | | | Lower | Upper | | | |
| Intercept[a] | 3.5840 | 8.6098 | -13.3355 | 20.50338 | 0.41627 | .677 | |
| debt_ratio_pct | -0.0354 | 0.0171 | -0.0690 | -0.00171 | -2.06527 | .039 | -0.09370 |
| revenue_growth_pct | 0.1934 | 0.0382 | 0.1182 | 0.26848 | 5.05751 | <.001 | 0.22448 |

[a] Represents reference level

Model Coefficients - margin_noinfl

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| ln_assets | 0.3833 | 0.3595 | -0.3232 | 1.08979 | 1.06616 | .287 | 0.05055 |
| sector_group: | | | | | | | |
| Consumer – Industrials & Energy | 0.0107 | 1.1562 | -2.2614 | 2.28278 | 0.00923 | .993 | 0.00110 |
| Finance – Industrials & Energy | 6.8506 | 1.4195 | 4.0610 | 9.64016 | 4.82588 | <.001 | 0.70602 |
| Healthcare – Industrials & Energy | 0.4025 | 1.4007 | -2.3500 | 3.15501 | 0.28737 | .774 | 0.04148 |
| Tech & Comms – Industrials & Energy | 2.6339 | 1.2567 | 0.1644 | 5.10347 | 2.09597 | .037 | 0.27146 |

[a] Represents reference level

## Data Summary

Cook's Distance

| | | | Range | |
|---|---|---|---|---|
| Mean | Median | SD | Min | Max |
| 0.00242 | 6.02e-4 | 0.00544 | 6.34e-9 | 0.0924 |

## Assumption Checks

Durbin–Watson Test for Autocorrelation

| Autocorrelation | DW Statistic | p |
|---|---|---|
| -0.0531 | 2.10 | .282 |

**[4]**

Collinearity Statistics

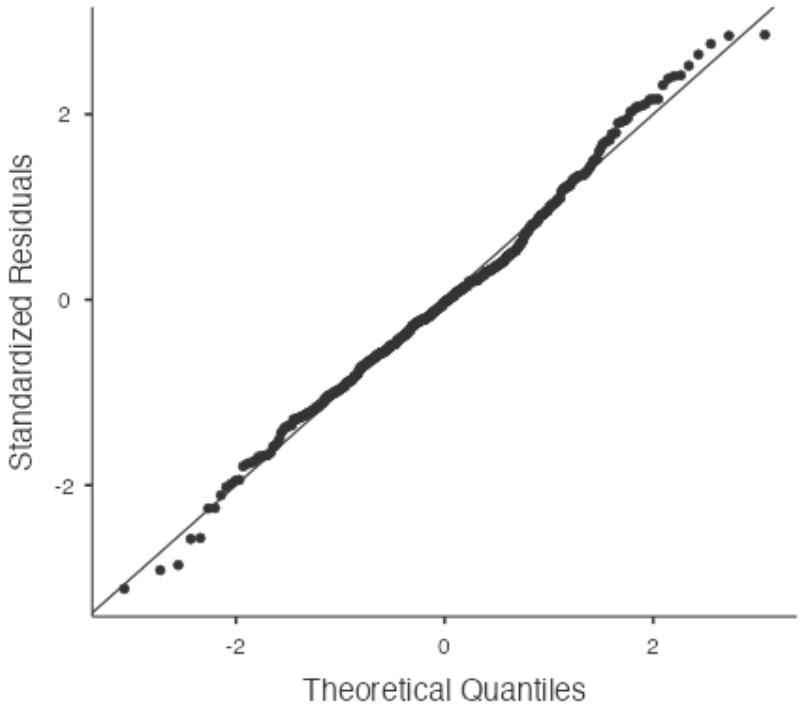
| | VIF | Tolerance |
|---|---|---|
| debt_ratio_pct | 1.05 | 0.952 |
| revenue_growth_pct | 1.03 | 0.973 |
| ln_assets | 1.10 | 0.911 |
| sector_group | 1.03 | 0.967 |

**[4]**

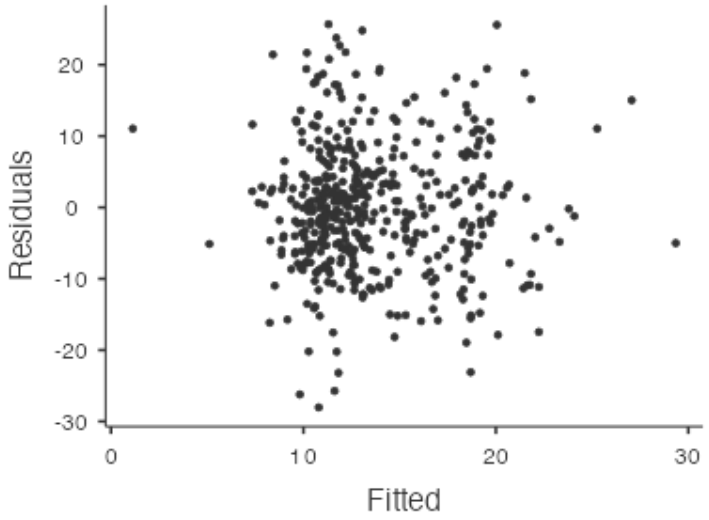
Normality Test (Shapiro-Wilk)

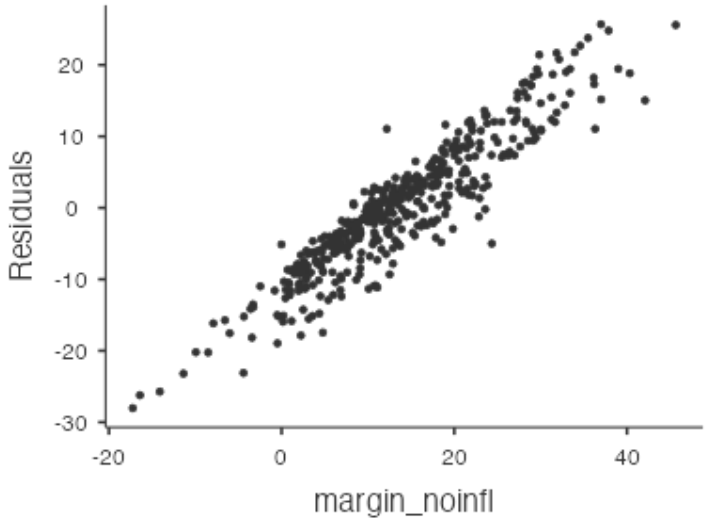
| Statistic | p |
|---|---|
| 0.992 | .014 |

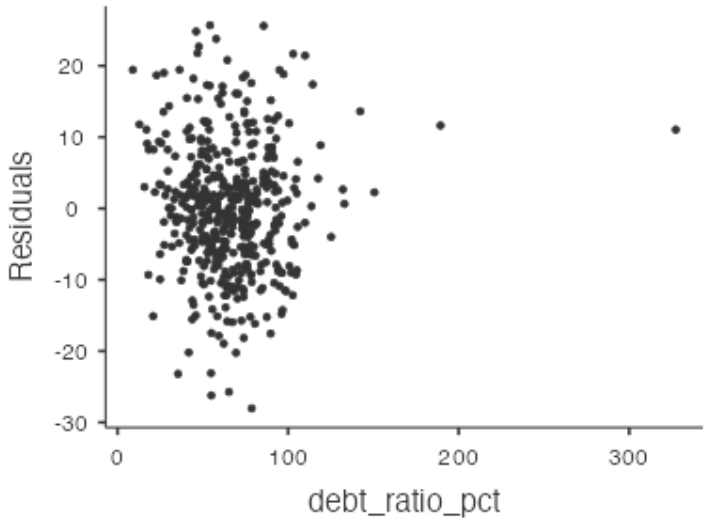
**Q-Q Plot**

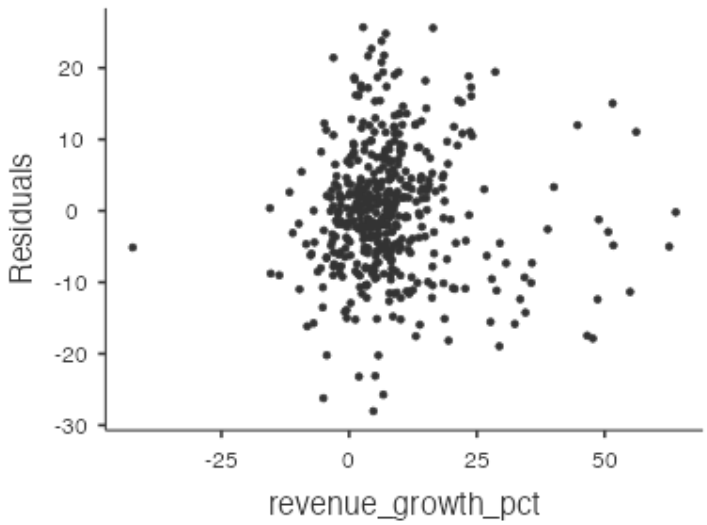


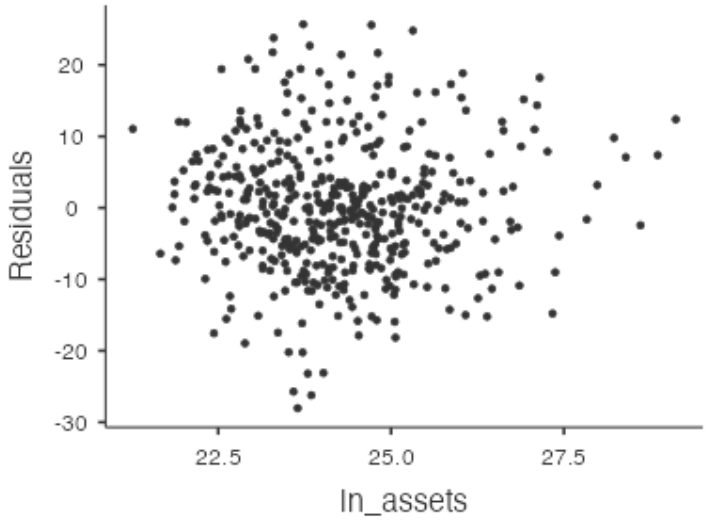

**Residuals Plots**




Residuals
20
10
0
-10
-20
-30
-20
0
20
40
margin_noinfl


Residuals
20
10
0
-10
-20
-30
0
100
200
300
debt_ratio_pct


Residuals
20
10
0
-10
-20
-30
-25
0
25
50
revenue_growth_pct




## Linear Regression

Model Fit Measures

| | | | | Overall Model Test | | | |
|---|---|---|---|---|---|---|---|
| Model | R | $R^2$ | Adjusted $R^2$ | F | df1 | df2 | p |
| 1 | 0.288 | 0.0829 | 0.0695 | 6.19 | 6 | 411 | <.001 |

Note. Models estimated using sample size of N=418

Model Coefficients - net_margin_pct

| | | | 95% Confidence Interval | | | | |
|---|---|---|---|---|---|---|---|
| Predictor | Estimate | SE | Lower | Upper | t | p | Stand. Estimate |
| Intercept[a] | 13.1012 | 15.3901 | -17.1520 | 43.3544 | 0.851 | .395 | |
| debt_ratio_pct | 0.0114 | 0.0248 | -0.0373 | 0.0602 | 0.461 | .645 | 0.0223 |
| revenue_growth_pct | 0.2500 | 0.0464 | 0.1589 | 0.3411 | 5.393 | <.001 | 0.2615 |
| ln_assets | -0.1601 | 0.6328 | -1.4041 | 1.0839 | -0.253 | .800 | -0.0120 |
| sector_nofin: | | | | | | | |
| Consumer – Industrials & Energy | 0.2735 | 1.8053 | -3.2753 | 3.8222 | 0.151 | .880 | 0.0183 |
| Healthcare – Industrials & Energy | -2.2482 | 2.1925 | -6.5580 | 2.0617 | -1.025 | .306 | -0.1506 |
| Tech & Comms – Industrials & Energy | 2.3553 | 1.9327 | -1.4439 | 6.1546 | 1.219 | .224 | 0.1578 |

[a] Represents reference level

## Data Summary

Cook's Distance

| | | | Range | |
|---|---|---|---|---|
| Mean | Median | SD | Min | Max |
| 0.00588 | 2.85e-4 | 0.0549 | 1.36e-10 | 0.870 |

## Assumption Checks

Durbin–Watson Test for Autocorrelation

| Autocorrelation | DW Statistic | p |
|---|---|---|
| 0.0697 | 1.86 | .134 |

**[4]**

Collinearity Statistics

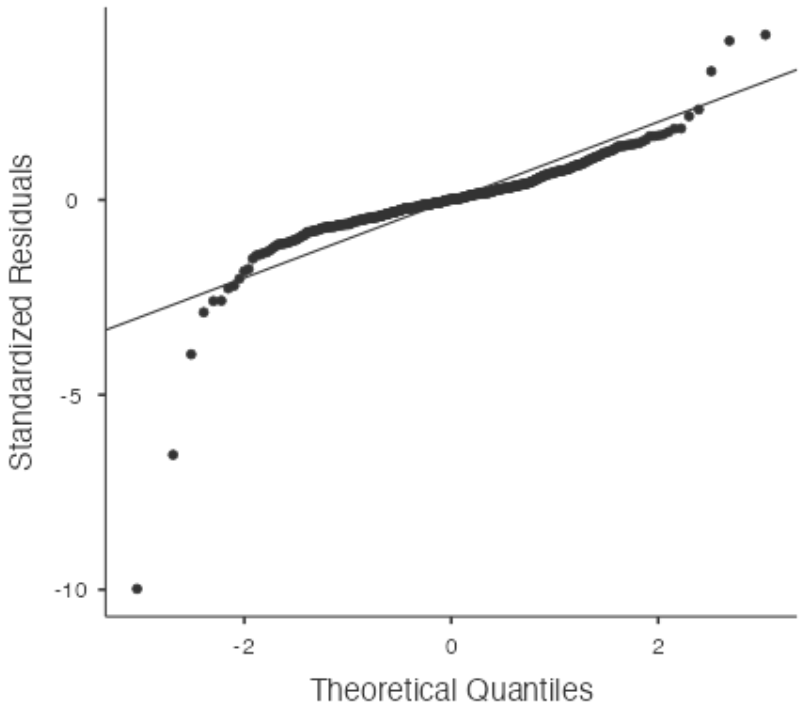
| | VIF | Tolerance |
|---|---|---|
| debt_ratio_pct | 1.02 | 0.979 |
| revenue_growth_pct | 1.03 | 0.974 |
| ln_assets | 1.00 | 0.997 |
| sector_nofin | 1.01 | 0.987 |

**[4]**

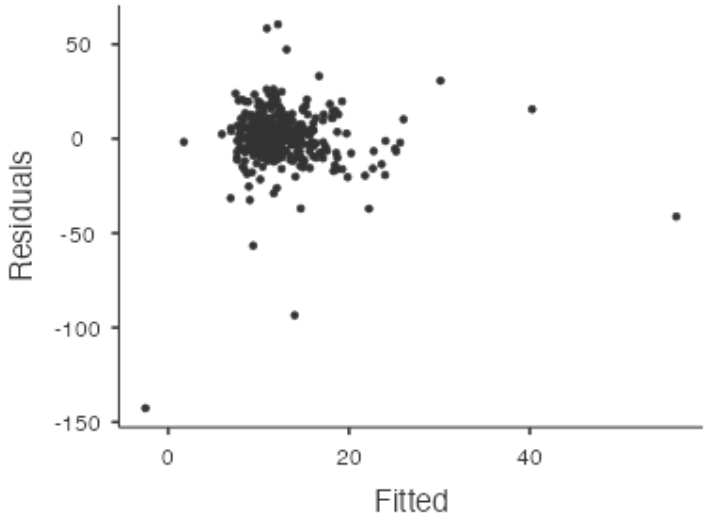
Normality Test (Shapiro-Wilk)

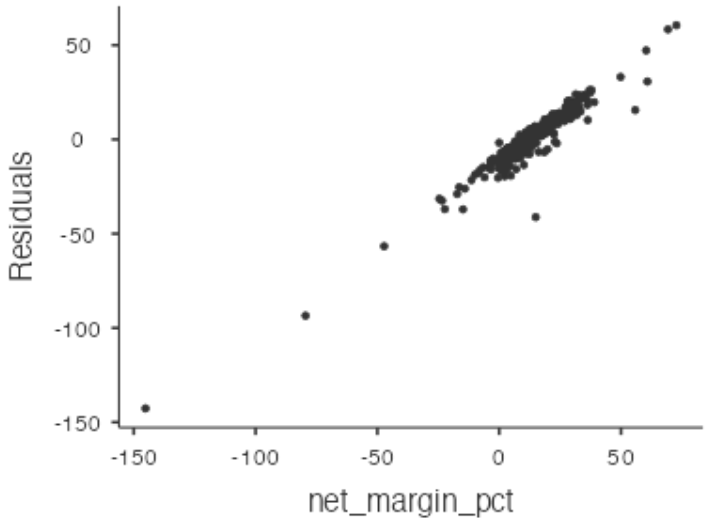
| Statistic | p |
|---|---|
| 0.776 | <.001 |

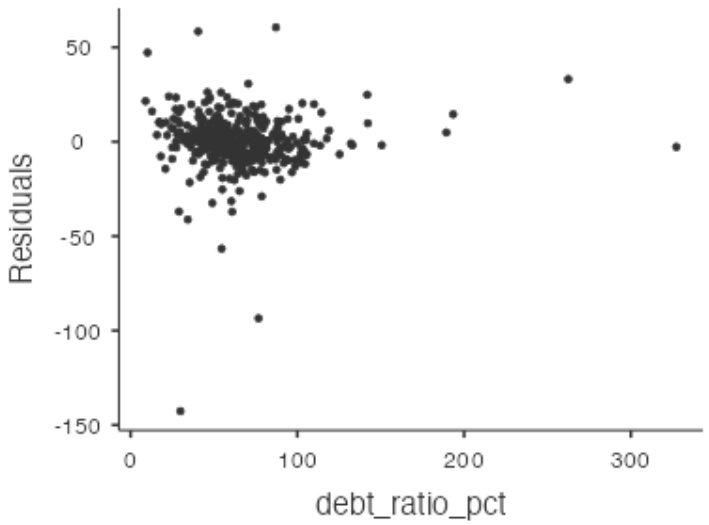
**Q-Q Plot**

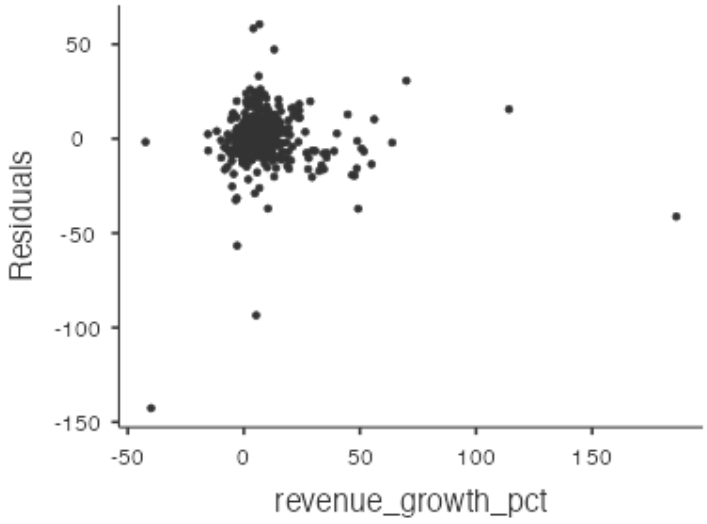


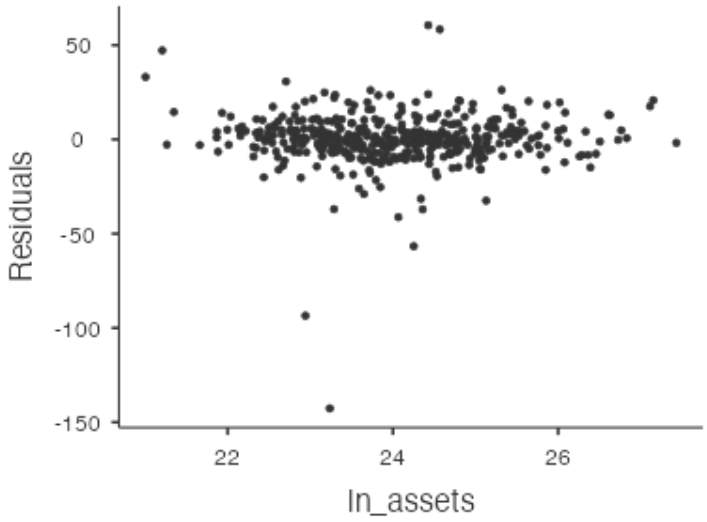

**Residuals Plots**


Residuals
50
0
-50
-100
-150
0
20
40
Fitted


Residuals
50
0
-50
-100
-150
-150
-100
-50
0
50
net_margin_pct


Residuals
50
0
-50
-100
-150
0
100
200
300
debt_ratio_pct




## References

**[1]** The jamovi project (2025). *jamovi*. (Version 2.7) [Computer Software]. Retrieved from https://www.jamovi.org.

**[2]** R Core Team (2025). *R: A Language and environment for statistical computing*. (Version 4.5) [Computer software]. Retrieved from https://cran.r-project.org. (R packages retrieved from CRAN snapshot 2025-05-25).

**[3]** Kerby, D. S. (2014). The simple difference formula: An approach to teaching nonparametric correlation. *Comprehensive Psychology. link, 3*, 2165–2228.

**[4]** Fox, J., & Weisberg, S. (2024). *car: Companion to Applied Regression*. [R package]. Retrieved from https://cran.r-project.org/package=car.